

PRIMARY AXIS

choose by optimization object

LOCAL

one serving instance / control domain

DISTRIBUTED

across devices, workers, replicas or nodes

execution scope expands across entities

SERVING POLICY & ROUTING

Decision object:
which model, worker or service path?

Single instance / local control domain

- Model / adapter / drafter selection
- Local cascade, fallback and path policy

Across replicas, pools, services or regions

- Replica / pool / region routing
- KV-aware routing and service orchestration

RUNTIME

Decision object:
online request and token execution

KV / Memory

Batch / Schedule

Decoding

Execution

KV / Memory

Paging, prefix reuse, eviction, offload

Batching / Scheduling

Continuous batch and chunked prefill

Decoding

Speculative, tree and parallel decoding

Execution

Graph replay and compute-copy overlap

KV / Memory

Distributed KV, migration, remote reuse

Batching / Scheduling

Cross-worker queues and PD pools

Decoding

Draft-target split and verification

Execution

TP / PP / EP and communication overlap

KERNEL

Decision object:
operator / communication primitive

Within one device / host

- FlashAttention, fused GEMM / RMSNorm
- Low-bit kernels, tiling, memory layout

Across devices

- All-reduce / all-to-all / P2P
- RDMA / NVLink paths, fused collectives

MODEL

Decision object:
parameters, structure and semantics

Model remains locally executable

- Quantization, pruning, distillation
- GQA / MQA / MLA, sparse attention

Architecture supports partitioned execution

- MoE / expert-friendly structure
- Partitionable attention, KV, model family

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